C20 Electrode Potentials and Cells – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Half-Cells and the SHE **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Standard electrode potential				EMF of a half-cell against the SHE under standard conditions
2	Conditions in the standard hydrogen electrode				H_2 at 100 kPa, 1.00 mol dm^{-3} H^+, 298 K
3	Electrode in the SHE				Platinum
4	$E^{\ominus}$ of the SHE				0.00 V
5	Ion concentration in a standard half-cell				1.00 mol dm^{-3}
6	Electrode used when no metal is in the half-cell				Platinum
7	Why platinum is used				Inert and conducts
8	Purpose of the salt bridge				Completes the circuit; lets ions move
9	Contents of the salt bridge				Saturated KNO_3
10	Why KCl is not used with Ag^+ half-cells				Forms AgCl precipitate

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Cells and Predictions **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	EMF equation				$E_{cell} = E_{right} - E_{left}$
2	Symbol for a phase boundary in cell notation				|
3	Symbol for the salt bridge in cell notation				||
4	Side of the cell diagram where oxidation occurs				Left
5	Cell notation for the Zn/Cu cell				$Zn(s)\|Zn^{2+}(aq)\|\|Cu^{2+}(aq)\|Cu(s)$
6	Half-cell that is the negative electrode				More negative $E^{\ominus}$
7	Strongest oxidising agent				Most positive $E^{\ominus}$ (left side)
8	Strongest reducing agent				Most negative $E^{\ominus}$ (right side)
9	Sign of E_{cell} for a feasible reaction				Positive
10	Why a feasible reaction may not happen				High E_a or non-standard conditions

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Commercial Cells **Paper 1**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Why some cells can be recharged				Reactions are reversible
2	Lithium-ion cell: negative electrode half-equation				$Li \rightarrow Li^+ + e^-$
3	Lithium-ion cell: positive electrode half-equation				$Li^+ + CoO_2 + e^- \rightarrow Li^+[CoO_2]^-$
4	Fuel cell				Cell supplied continuously with fuel and oxygen
5	Alkaline hydrogen fuel cell: negative electrode				$H_2 + 2OH^- \rightarrow 2H_2O + 2e^-$
6	Alkaline hydrogen fuel cell: positive electrode				$O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$
7	Overall fuel cell equation				$2H_2 + O_2 \rightarrow 2H_2O$
8	Advantage of hydrogen fuel cells				Only product is water
9	Disadvantage of hydrogen fuel cells				H_2 is flammable, hard to store, often made from fossil fuels
10	Why fuel cells do not need recharging				Reactants supplied continuously

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.